

Contribution Journal: Mid-Term Project

Course: ITAI 1371 - Introduction to Machine Learning

Group: ML_1371_12321 Group 4

Date: July 5, 2026

Group Members

- Denis Aosa (Denis Bosire)
 - Aven Gilbert
 - Victoria Simmons (Tori)
 - Mary Balemba
 - Raymond Hayes
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Individual Contributions for Mid-Term Project

Denis Aosa — Contributions

- **Jupyter Notebook Implementation:** Manually completed and executed all 5 major preprocessing tasks in the joint notebook `app.ipynb` (Tasks 1-5):
- **Task 1:** Implemented one-hot encoding on categorical variables (`room_type`, `cancellation_policy`) with `drop_first=True`.
- **Task 2:** Balanced target variable classes (`instant_bookable`) via majority class downsampling without replacement using `sklearn.utils.resample`.
- **Task 3:** Converted categorical variables (`neighbourhood_group`, `host_identity_verified`) to category codes.
- **Task 4:** Cleaned features and dropped non-predictive metadata columns to export the final preprocessed dataset (`Airbnb_Cleaned_Data.csv`).
- **Task 5:** Implemented Scikit-Learn's `train_test_split` to divide the dataset into 70% training and 30% testing.
- **Documentation & Proposal:** Drafted the dataset description and project proposal document.
- **Reflective Journal:** Authored individual reflective journal detailing learning outcomes.

Aven Gilbert — Contributions

- **Project Setup:** Initialized the GitHub repository and uploaded raw dataset.

- **Data Integration:** Created the base skeleton of `app.ipynb` and defined the pre-processing tasks and TODO structure.
- **Visualizations:** Added initial visualization code cells and data distributions in the notebook.

Victoria Simmons, Mary Balemba, Raymond Hayes — Contributions

- **Code Review & Feedback:** Reviewed and validated the pre-processing logic in the notebook.
- **Project Discussions:** Participated in dataset selection and target variable confirmation discussions.
- **Documentation:** Contributed individual reflective notes and finalized submission components.